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EE/RPF

For use from May/November 2027

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Student personal code:

Extended essay - Reflection and progress form

Student: *This form is to be completed by you and your supervisor during the progress and completion of the extended essay. It records your progress through the extended essay process and includes your reflective statement addressing your extended essay experience. You must undertake three formal reflection sessions with your supervisor: The first formal reflection session should focus on your initial ideas and how you plan to undertake your research; the interim reflection session takes place once a significant amount of your research has been completed, and the final session will be in the form of a viva voce, once you have completed and handed in your extended essay. This document acts as a record in supporting the authenticity of your work. Following the viva voce, you write a* ***reflective statement*** *of no more than 500 words., which must be included on the RPF.*

The completion of this form is a mandatory requirement of the extended essay. It must be submitted together with the completed essay. The reflective statement will then be assessed under Criterion E. Students and supervisors should be aware that a mark of 0 will be awarded by the examiner for criterion E if the RPF is blank or the reflective statement is written in a language other than that of the accompanying essay.

Supervisor: *You must have three reflection sessions with each student, one early in the process, an interim meeting and then the final viva voce. Other check-in sessions are permitted but do not need to be recorded on this sheet. After each reflection session you must record the month that the reflection session took place, the student's year of DP study at that time and initial the form.*
By submitting this student work for assessment, you are taking responsibility for its authenticity. No piece of student work should be uploaded/submitted to the e-Coursework system if its authenticity is in doubt.

First reflection session

Month: DP year (1 or 2): Supervisor initials:

Interim reflection session

Month: DP year (1 or 2): Supervisor initials:

Final reflection session – Viva voce

Month: DP year (1 or 2): Supervisor initials:

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Reflective statement

(No more than 500 words, written in the same language as the accompanying extended essay)

As a Physics student, I am most attracted to mechanics. Once, when I was discussing with my dad about the possible time required for a rocket to travel to the Moon, I noticed that craters on the Moon's surface had very different sizes. I was intrigued and decided to investigate the mechanics behind it. My initial thoughts were to simulate the situation by dropping marbles on flour. However, I realized that the crater created this way was way too small and its volume hard to measure. Hence, I decided to use larger objects such as metal balls and Gashapon shells with weights. In this way I was able to plan and conduct a smooth experiment. Planning was the first key stage that allowed me to organize and envision possible scenarios, planning that I then copied for other tasks in other subjects.

During the experiment I encountered multiple challenges that I had not considered. First, I realized that a crater cannot be formed as the flour is too loose. Secondly, my original metal ball was way too heavy and the relationship between the height of the ball and the volume of the crater can hardly be identified clearly. Although I do not have a lot of tools, I was able to solve these issues with simple assets. To tighten up the flour bed, I compressed it with a box. The ball was also changed to a table tennis ball filled with water to vary its mass while keeping the volume constant. I also found that my original way of measuring the volume of the crater was too hard and led to inaccuracies due to systematic errors. Hence, I decided to estimate the volumes using their diameters instead to lower the difficulty.

The process has been very valuable. Throughout the investigation, I have done substantial research, developing skills that I have applied as well into other subjects. Each mistake I made has taught me something that I had never noticed before in designing an experiment. For example, simply finding a suitable ball was already a challenge as I needed one moderate in size with a low density such that it would not drop destroying the set up. There were also other significant systematic inaccuracies that I had to modify and deal with to obtain an accurate result, such as the density of the flour. In the end, although I am satisfied by the result of the investigation, it is encouraging to me and would motivate me to do other studies in the future. I would also like to redo these experiments with more detailed measurements of the crater to look for a more significant and convincing relationship.

The final process of writing and reviewing mentally all the long process was also rewarding and induced me to recap and value each step.

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